

LESSON 6-3 • SESSION 1 • PRACTICE • FORM B

Writing Repeated Factors as Powers

Learning target – I can write repeated multiplication as a power and name its base and exponent. • 6.EE.A.1

REMEMBER

- A **power** shows a repeated factor: $5 \times 5 \times 5 = 5^3$.
- The **base** is the repeated factor. The **exponent** tells how many times it is a factor.
- Read 5^3 as "5 to the third power" or "5 cubed." 5^2 is "5 squared."

WORKED EXAMPLE

Write $9 \times 9 \times 9 \times 9$ as a power.**Step 1** The repeated factor is 9, so the base is 9.**Step 2** Count the 9s: there are four, so the exponent is 4.**Step 3** Check: the exponent counts factors, not $\times$ signs (there are only 3 $\times$ signs).**Answer** 9^4 ; base 9, exponent 4**WORDS** **power** (*potencia*) • **base** (*base*) • **exponent** (*exponente*)

PRACTICE

show your work • label units

1. Write each product as a power.

a. $6 \times 6 \times 6 =$ _____

b. $9 \times 9 \times 9 \times 9 =$ _____

c. $11 \times 11 =$ _____

d. $5 \times 5 \times 5 \times 5 \times 5 =$ _____

e. $2/3 \times 2/3 \times 2/3 =$ _____

f. $0.7 \times 0.7 =$ _____

- 2.
- Seedlings and Garden.**
- A greenhouse's seedling count is multiplied by
- $3 \times 3 \times 3 \times 3 \times 3$
- . A square garden has
- 12×12
- plants.

a. Write the seedling product as a power. _____

b. Write the garden product as a power. _____

c. In the seedling power, name the base and the exponent. _____

d. What does the exponent count? _____

- 3.
- Game Rounds.**
- A score is multiplied by 4 in each of six rounds:
- $4 \times 4 \times 4 \times 4 \times 4 \times 4$
- . Write this as a power. Name the base and exponent.

Hint The repeated factor is the base.

Show your work.

Power _____

- 4.
- Ant Colony.**
- A science report records an ant count as
- $15 \times 15 \times 15 \times 15$
- . Write this as a power. Name the base and exponent.

Hint Count the factors of 15.

Show your work.

Power _____

- 5.
- Block Tower.**
- Lena wrote the product
- $6 \times 6 \times 6 \times 6$
- as the power
- 6^6
- . Justify or correct her work.

Hint What does the exponent count? Recount the factors.**Power** _____

6. **Two Directions.** Write each one the other way.

Hint Put parentheses around a fraction base.

- a. Write $1/5 \times 1/5$ as a power. _____
- b. Write 20^3 as a product of factors. _____

Show your work.

7. **Growing Bacteria.** In each growth period, a bacteria count triples. After 1 period there are 3 bacteria.

- a. Write the number of bacteria after 4 periods as a power of 3. _____
- b. Find the number of bacteria after 4 periods. _____ bacteria
- c. Explain how each new period changes the power. _____

8. **Swapped Numbers.** Compare 2^5 and 5^2 .

- a. Write each power in expanded form. _____
- b. Which power has more factors? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Imani's Work.** Imani read the power 3^5 and wrote: " $3 \times 5 = 15$, so 3^5 equals 15." Explain her mistake. Then write 3^5 correctly as a product.

TEST PRACTICE

choose the best answer

10. In 8^3 , the exponent is –

- A 8 B 3 C 24 D 11

11. The exponent tells you how many –

- A times the base is used as a factor B times to add the base
- C digits are in the value D times to multiply by 10

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

81 L	3 ⁵ M	15 ⁴ L	2 ⁵ Y	4 ⁶ U	(1/5) ² I	6 ⁴ T	12 Z	3 ⁴ P	6 ⁶ Q
#2a	#3	#4	#5	#6a	#7a	#7b	#8b		

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____